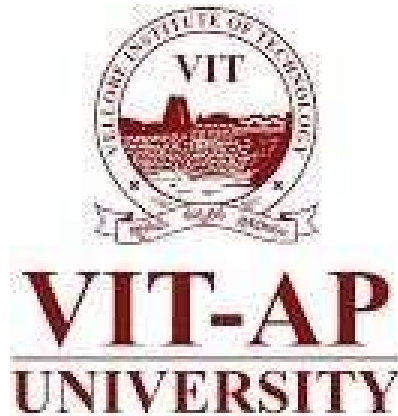




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Convulsive Seizure Detection and Notification Device

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ABSTRACT:

This project focuses on the development of a wearable device for convulsive seizure detection and notification, aimed at providing timely alerts to improve safety and response for individuals with epilepsy. The device is built using a Raspberry Pi, an ADXL335 accelerometer, an Arduino Nano/UNO, and an HC-05 Bluetooth module. It continuously monitors body movements and detects seizure-like motion patterns, utilizing accelerometer data and proprietary algorithms to accurately identify seizure events.

The device operates in real-time, sending notifications to designated contacts when a seizure is detected. This allows for rapid intervention, which is critical for reducing the risks associated with unattended seizures, such as injuries and prolonged medical response times. The system is developed using open-source tools and programming languages, including Python, with custom algorithms designed to minimize false positives and improve detection accuracy.

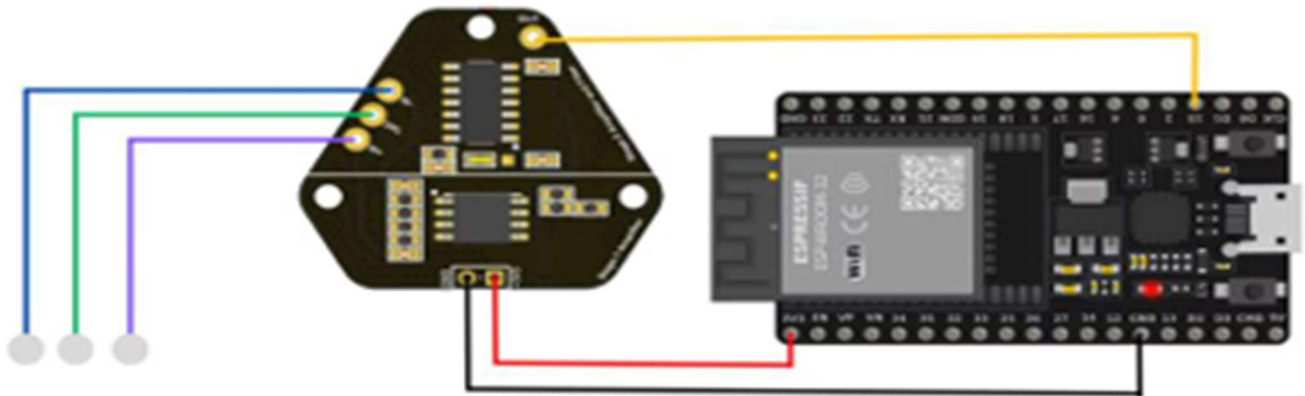
To ensure user convenience and efficiency, the device is optimized for power management, allowing extended monitoring periods on a single battery charge. The project also addresses challenges in data interpretation, sensitivity calibration, and device wearability to enhance user comfort and reliability.

This wearable seizure detection device has considerable potential for further enhancement. Integrating AI algorithms in future iterations could improve the device's ability to differentiate between seizure and non-seizure movements, thus increasing its reliability. The system can also be expanded with additional sensors, such as heart rate monitors or environmental sensors, to provide a more comprehensive health monitoring solution for individuals with epilepsy. By prioritizing user privacy and adherence to regulatory standards, this device serves as a vital step towards improving the quality of life and safety for those affected by epilepsy.

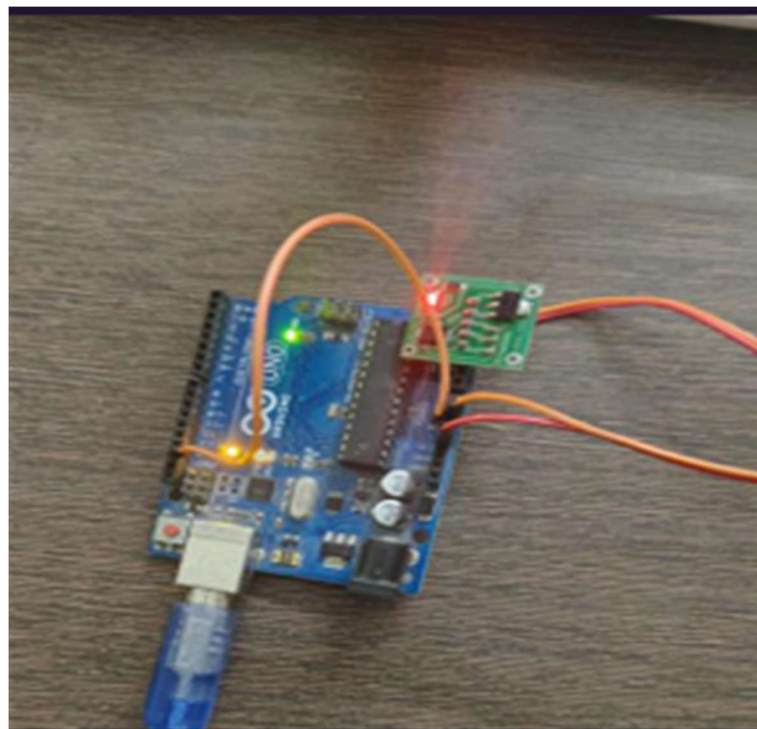
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Circuit Diagram



Hardware Configuration

INTRODUCTION:

In today's world, technology continuously evolves to address various needs, making daily tasks simpler and enhancing quality of life. One such critical area where technology has a transformative impact is in health monitoring, particularly for individuals managing chronic conditions like epilepsy. Convulsive seizures, common in epilepsy, can occur unpredictably and pose significant risks, including injury and delayed medical intervention if left unmonitored. Conventional methods for seizure detection are often intrusive, uncomfortable, or require constant supervision, limiting their practical application.

This project seeks to address these challenges by developing a wearable, real-time convulsive seizure detection and notification device. Designed to be non-intrusive, this device continuously monitors the user's movements, utilizing an ADXL335 accelerometer and an Arduino Nano/UNO to detect sudden, seizure-like motion patterns. Upon detecting such patterns, the device triggers an alert, immediately notifying designated contacts through the HC-05 Bluetooth module, ensuring that assistance can be reached promptly. This innovation is crucial for individuals with epilepsy, as timely intervention can prevent injuries and provide peace of mind for both users and caregivers.

The device is portable, powered by a 3.7v battery, and optimized for continuous use, making it suitable for home, work, and other environments. By harnessing modern hardware and algorithmic capabilities, this seizure detection device not only enhances user safety but also opens new possibilities for future advancements, such as integrating additional sensors or AI algorithms to improve accuracy. Ultimately, this project aims to improve the quality of life for individuals with epilepsy by providing a reliable, proactive solution for convulsive seizure management.

BACKGROUND:

Seizure monitoring and management have traditionally relied on self-reporting, caregiver observation, or wearable devices that can be intrusive and uncomfortable. In conventional methods, patients may wear EEG headgear or heart rate monitors, which can be restrictive and inconvenient, especially in daily life settings. Furthermore, these devices often require constant supervision or manual intervention to recognize seizure events, limiting their practicality for continuous, real-world use.

With the advancement of wearable technology and real-time processing, non-intrusive devices are now feasible for tracking and identifying seizure-like movements. By utilizing low-power components and compact sensors, modern devices can provide accurate monitoring without disrupting the user's daily activities. The ADXL335 accelerometer, for instance, is a small, 3-axis sensor capable of detecting even subtle changes in motion, making it suitable for seizure detection. Combined with the processing power of Arduino Nano/UNO and the connectivity enabled by Bluetooth modules like the HC-05, these components create a wearable solution that is portable, responsive, and capable of delivering immediate alerts.

This project harnesses this technology, using a combination of an accelerometer and microcontroller to develop a reliable, real-time seizure detection system. Designed with the user's comfort in mind, the device allows for continuous monitoring and rapid notification during seizure events, offering an innovative approach to improve safety, independence, and peace of mind for individuals with epilepsy and their caregivers.

PROBLEM DEFINITION:

The problem addressed by this project is the need for an efficient and non-intrusive seizure detection system for individuals with epilepsy, providing timely alerts to caregivers or medical professionals during a seizure event.

Seizures, particularly convulsive ones, are unpredictable and can result in serious injuries if not managed immediately. Current methods of seizure detection, such as manual observation or invasive EEG monitoring, can be unreliable, uncomfortable, and impractical for continuous use. There is a need for a portable, user-friendly device that can accurately detect convulsive seizures and alert caregivers or medical personnel in real-time without causing discomfort to the patient.

The proposed system uses an ADXL335 accelerometer, an Arduino-based processing unit, and Bluetooth technology to detect specific movement patterns associated with convulsive seizures. The accelerometer continuously monitors the user's body movements and processes this data in real time to identify patterns indicative of a seizure. If a seizure is detected, the system immediately triggers an alert, sending notifications to pre-defined contacts (such as caregivers, family members, or healthcare professionals) to ensure timely intervention.

This system will provide a non-invasive, accurate, and reliable way to monitor patients with epilepsy, offering peace of mind for caregivers and improving overall patient safety. The proposed seizure detection device can be worn comfortably throughout the day, providing continuous monitoring without disruption to the user's lifestyle, making it a more feasible and effective solution compared to existing methods.

OBJECTIVES:

The proposed convulsive seizure detection and notification system aims to achieve the following objectives:

- **To design and implement a wearable hardware system** that includes an ADXL335 accelerometer, an Arduino processing unit, and Bluetooth technology for detecting convulsive seizures based on motion patterns.
- **To develop and integrate proprietary algorithms** for real-time analysis of accelerometer data to accurately identify seizure events and distinguish them from normal activities such as exercise or minor movements.
- **To create a notification system** that alerts caregivers, family members, or healthcare professionals immediately upon detecting a seizure, using methods such as SMS, email, or mobile app notifications.
- **To ensure comfort and reliability** by designing a lightweight, user-friendly device that can be worn throughout the day without causing discomfort, while maintaining continuous monitoring and alerting functionality.
- **To test the system in various environments** and under different conditions to evaluate its performance, including the accuracy of seizure detection, the reliability of alerts, and the device's battery life.
- **To optimize the system's performance** by refining the algorithms and hardware components based on test results to minimize false positives and enhance overall reliability.
- **To provide a cost-effective, efficient, and practical solution** for seizure detection that improves patient safety and provides peace of mind for caregivers, with potential deployment in residential homes, hospitals, and other healthcare settings.

PROCEDURE:

COMPONENTS REQUIRED:

- **ADXL335 Accelerometer** – A 3-axis accelerometer to measure acceleration along the X, Y, and Z axes, enabling motion detection for convulsive seizures.
- **Arduino Nano/UNO** – A microcontroller used for processing the data from the accelerometer and controlling the system's operations, including triggering notifications and alarms.
- **HC-05 Bluetooth Module** – Used for wireless communication, allowing the system to send alerts and notifications to designated contacts via mobile phones or other devices.
- **Buzzer** – A sound alarm triggered when a seizure is detected, alerting caregivers or others nearby to the event.
- **3.7V Lithium Battery** – A compact power supply to ensure the device remains functional for extended periods without constant recharging.
- **Connecting Wires** – For connecting components such as the accelerometer, Arduino, and Bluetooth module to ensure proper data transmission and functionality.
- **USB Cable** – Used for powering the Arduino and for uploading code to the microcontroller.
- **Proprietary Algorithms** – Software algorithms implemented in the Arduino to process the accelerometer's data and accurately detect seizure patterns.
- **Enclosure** – A lightweight, comfortable case or strap that houses the components and ensures easy wearability while maintaining functionality.

The Convulsive Seizure Detection and Notification Device is an embedded system developed using the Raspberry Pi platform. It integrates an ADXL335 accelerometer, Arduino, Bluetooth module, and a buzzer to monitor and detect convulsive seizures. The accelerometer continuously tracks motion in three axes and sends this data to the Arduino, which processes the signals and detects any seizure-like activity. Upon detecting a seizure, the system triggers the buzzer, displays the message “SEIZURE DETECTED” on the LCD screen, and sends notifications to designated contacts via the HC-05 Bluetooth module. Additionally, images or data related to the seizure event are sent to the designated contacts for immediate intervention. This device provides real-time alerts to caregivers, improving response times and enhancing the safety of individuals with epilepsy.

Arduino:

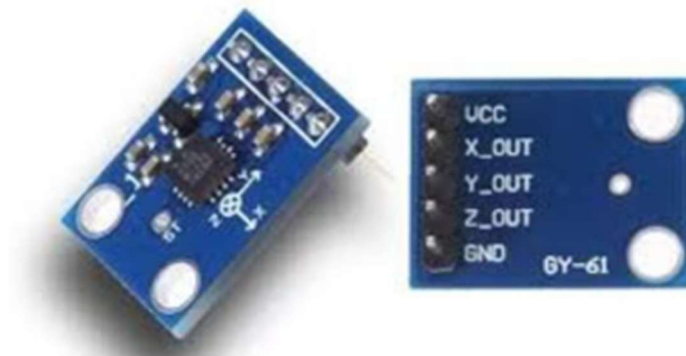


Arduino is an open-source microcontroller platform that provides an easy-to-use environment for developing interactive electronic projects. It consists of both hardware (Arduino boards) and software (Arduino IDE), enabling users to design and implement various embedded systems. Arduino boards are based on a microcontroller (often an ATmega series processor) and come with various input and output pins, such as digital, analog, PWM, and serial communication ports, which allow for connecting sensors, actuators, and other peripherals.

Arduino is highly popular due to its simplicity and versatility, making it accessible to both beginners and professionals. The platform supports multiple programming languages, with the most common being C and C++, and the Arduino IDE allows for easy code writing, compilation, and uploading to the board. It also supports numerous development environments, libraries, and third-party add-ons to extend its capabilities.

Arduino is widely used in a variety of applications, including robotics, home automation, environmental monitoring, and IoT projects. Its open-source nature, extensive community support, and low cost make it a go-to solution for prototyping, educational purposes, and hobbyist projects.

ADXL335 Accelerometer:

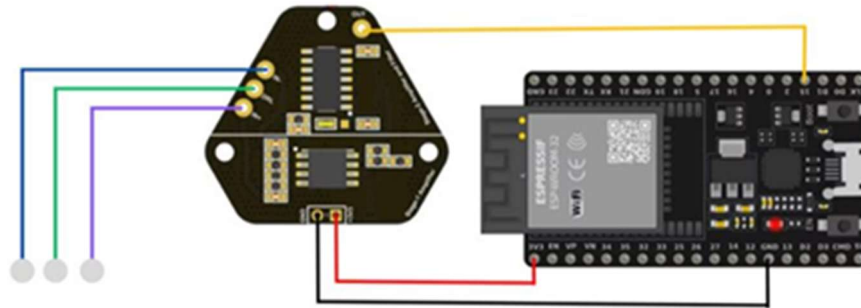


The ADXL335 accelerometer is a small, low-power, 3-axis analog accelerometer used to measure acceleration forces in the X, Y, and Z axes. It is capable of detecting both static forces, like gravity, and dynamic forces, such as movement or vibration. The ADXL335 provides analog voltage outputs corresponding to the acceleration along each of the three axes, making it suitable for applications where precise motion and orientation detection are needed.

The ADXL335 is widely used in various applications, including motion detection, tilt sensing, vibration analysis, and free-fall detection. It is commonly used in robotics, wearable devices, and mobile applications to track movement, orientation, or position. Its small form factor and low power consumption make it ideal for embedded systems and battery-powered projects.

With its high sensitivity and ability to detect subtle changes in acceleration, the ADXL335 is a versatile sensor that can be integrated into a wide range of devices and systems for motion tracking, navigation, and environmental monitoring.

EXG sensor:



The EXG (Electromyography) sensor is a specialized device used to measure the electrical activity produced by skeletal muscles. It detects muscle contractions by capturing the electrical signals generated during muscle activation. The EXG sensor works by amplifying and filtering the weak electrical signals produced by muscles, converting them into a readable signal for further analysis or processing.

EXG sensors are widely used in medical and healthcare applications, such as for monitoring muscle activity in patients with neurological disorders or for rehabilitation purposes. They are also used in research, prosthetics, and human-computer interaction (HCI) systems, where muscle movements are translated into control signals for devices or interfaces.

These sensors typically provide analog outputs that represent the intensity of muscle contractions, and they can be integrated into wearable devices or systems for real-time muscle monitoring. With the growing interest in biofeedback and wearable health technology, the EXG sensor has become a valuable tool in various applications that require the tracking of muscle performance, fatigue, and rehabilitation progress.

RESULTS AND DISCUSSION:

- **Improved Security:** The convulsive seizure detection and notification device enhances safety for individuals prone to seizures by providing real-time alerts. It monitors motion using the ADXL335 accelerometer and immediately detects any abnormal activity, such as convulsive seizures. When a seizure is detected, the device sends timely notifications to designated contacts via Bluetooth, ensuring a swift emergency response and reducing the risks associated with delayed intervention. This system significantly contributes to improving patient safety and minimizing the risk of injury during a seizure.
- **Enhanced Quality of Life:** By offering peace of mind to both patients and caregivers, this device significantly improves the quality of life. With the ability to continuously monitor a patient's movements and detect convulsions, the device ensures timely intervention, which is critical in preventing injury and other complications. It helps caregivers remain alert to the patient's needs without constant supervision, promoting autonomy and independence for the user.
- **Data Storage and Analysis:** The device collects data continuously, which can be useful for post-event analysis. Storing sensor data allows healthcare professionals to monitor the frequency and intensity of seizures, improving long-term care management. However, managing this data effectively while ensuring privacy and security is crucial. Proper encryption and secure storage practices must be implemented to protect sensitive health information.
- **Privacy Concerns:** As with any device that collects health data, privacy is a major concern. The information captured by the device, including seizure events and personal health data, should be handled in compliance with data protection regulations, such as HIPAA (Health Insurance Portability and Accountability Act) or GDPR (General Data Protection Regulation). Ensuring that only authorized individuals can access the data is essential to maintaining user trust.
- **Technical Challenges:** Building a reliable and efficient seizure detection system with minimal false positives and false negatives is a technical challenge. Ensuring the accuracy of the ADXL335 accelerometer, optimizing the algorithms for motion detection, and fine-tuning the communication between the hardware and software components are crucial for the system's success. Additionally, power management and ensuring the device's long battery life while maintaining high accuracy is an ongoing challenge. Ensuring the device is comfortable and wearable for users is also critical to its acceptance and use in everyday life.

In conclusion, this project demonstrates the potential of using wearable technology for improving safety and quality of life for people with epilepsy. While there are technical and privacy challenges to address, the device offers a promising solution for timely seizure detection and intervention.

CONCLUSION:

The convulsive seizure detection and notification device, built on the Raspberry Pi platform, presents a promising solution to enhance the safety and quality of life for individuals with epilepsy. By utilizing the ADXL335 accelerometer for motion detection, the system can accurately identify convulsive seizures in real-time and send instant alerts to designated contacts via Bluetooth. This ensures a prompt response, potentially preventing injuries and providing peace of mind to both the individuals experiencing seizures and their caregivers.

The device demonstrates significant potential for improving the autonomy of people with epilepsy, as it allows for continuous monitoring without the need for constant supervision. Additionally, the collected data provides valuable insights for healthcare professionals, contributing to better care management and long-term health tracking.

While there are challenges related to ensuring data privacy, optimizing device performance, and minimizing false positives, the system shows great promise for future development. By integrating more advanced sensors, refining algorithms, and expanding communication capabilities, this device can play a crucial role in the safety and well-being of individuals with epilepsy, offering both technological innovation and practical benefits.

Future Scope:

- **AI and Machine Learning Integration:** The convulsive seizure detection and notification device can benefit from the integration of advanced AI and machine learning algorithms to improve the accuracy of seizure detection. AI can be used to distinguish between different types of movements, reducing false positives and enhancing the system's ability to detect subtle seizure patterns that may not be easily identified through simple motion detection.
- **Integration with Health Monitoring Systems:** The system can be further expanded by integrating with wearable health monitoring devices, such as heart rate monitors or oxygen saturation sensors. This integration would provide a more comprehensive monitoring solution, allowing healthcare professionals to assess the individual's overall health status in real-time.
- **Cloud-Based Data Storage and Analytics:** Storing the seizure data on a cloud platform would enable better data management and analysis. Cloud-based analytics could be used to track trends over time, provide detailed reports, and offer insights into seizure patterns, which could aid in medical research and improve patient care.
- **Mobile App and Remote Monitoring:** The development of a mobile app for caregivers and medical professionals could allow for real-time notifications and remote monitoring of the user's condition. This would enable caregivers to respond quickly to alerts and access historical data for better decision-making.
- **Integration with Smart Home Systems:** The device could be integrated with smart home technologies such as home automation systems, allowing it to trigger responses such as dimming lights or activating alarms in the event of a seizure. This would further improve the overall safety and security of the user's environment.
- **Battery Life and Portability Enhancements:** Further research into low-power components and energy-efficient designs could significantly enhance the battery life and portability of the device, ensuring that it is suitable for long-term, continuous use without frequent recharging.

By addressing these areas, the convulsive seizure detection and notification device can evolve into a more robust and versatile solution, offering enhanced safety, better care, and valuable data for individuals with epilepsy and their support network.

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- [2] https://youtu.be/VCdTn4_ELA4
- [3] <http://synopsis.nevemtech.com/index>
- [4] <https://www.researchgate.net/>
- [5] <https://www.ids-securityltd.co.uk>
- [6] <https://thewire.in/rights>
- [7] <https://ijarcce.com>
- [8] <https://www.amazon.in/>
- [9] <https://www.canva.com/>

APPENDIX

CODE:

```
const int ap1 = A5;
const int ap2 = A4;
const int ap3 = A3;

int sv1 = 0;
int ov1 = 0;
int sv2 = 0;
int ov2= 0;
int sv3 = 0;
int ov3= 0;

void setup() {
  Serial.begin(9600);
}
```

```

void loop() {
    analogReference(EXTERNAL);
    sv1 = analogRead(ap1);
    ov1 = map(sv1, 0, 1023, 0, 255);
    delay(2);
    sv2 = analogRead(ap2);
    ov2 = map(sv2, 0, 1023, 0, 255);
    delay(2);
    sv3 = analogRead(ap3);
    ov3 = map(sv3, 0, 1023, 0, 255);

    Serial.print("Xsensor1 = ");
    Serial.print(sv1);
    Serial.print("\t output1 = ");
    Serial.println(ov1);

    Serial.print("Ysensor2 = ");
    Serial.print(sv2);
    Serial.print("\t output2 = ");
    Serial.println(ov2);

    Serial.print("Zsensor3 = ");
    Serial.print(sv3);
    Serial.print("\t output3 = ");
    Serial.println(ov3);

    delay(300);
}

```

Output :-

The output data comes from an ADXL335 accelerometer used in a **Convulsive Seizure Detection Device**. The sensor captures movement along the X, Y, and Z axes, and the device processes these values to detect abnormal patterns associated with convulsive seizures. If rapid, repetitive movements occur, the device triggers an alert, ensuring timely notification to caregivers for immediate medical intervention.

```
Xsensor1 = 228    output1 = 56
Ysensor2 = 242    output2 = 60
Zsensor3 = 283    output3 = 70
Xsensor1 = 229    output1 = 57
Ysensor2 = 241    output2 = 60
Zsensor3 = 282    output3 = 70
Xsensor1 = 229    output1 = 57
Ysensor2 = 242    output2 = 60
Zsensor3 = 283    output3 = 70
Xsensor1 = 229    output1 = 57
Ysensor2 = 242    output2 = 60
Zsensor3 = 283    output3 = 70
Xsensor1 = 229    output1 = 57
Ysensor2 = 242    output2 = 60
Zsensor3 = 282    output3 = 70
Xsensor1 = 230    output1 = 57
Ysensor2 = 242    output2 = 60
Zsensor3 = 284    output3 = 70
```

Code:-

```
#include <synapse.h>
// #include "../synapse.h"

const int sensorPin = A0; // Analog input pin
unsigned long previousMillis = 0; // Variable to store the last time a sample was taken
unsigned long interval = 4; // Interval between samples in milliseconds (1000 ms / 250 samples). Change as per requirement

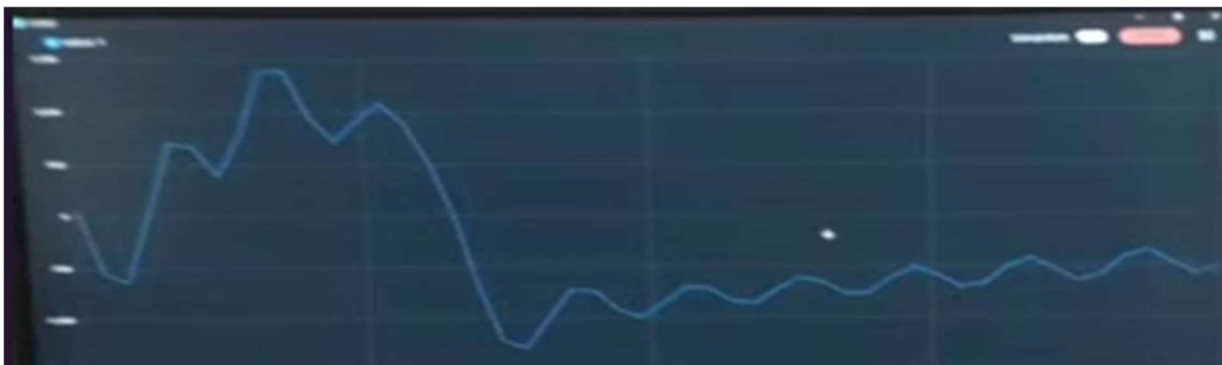
synapse exg_synapse;

void setup(){
  Serial.begin(115200);
}

void loop(){
  unsigned long currentMillis = millis(); // Get the current time in milliseconds

  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis; // Save the current time as the last time a sample was taken
    float sensorValue = analogRead(sensorPin); // Read the sensor value
    float filters_out = exg_synapse.apply_EEG_filters(sensorValue); // generate output after applying filters
    Serial.println(filters_out);
    delay(500);
  }
}
```

Output (Meditative State)



Output (Active State)

